

Leveraging LLMs for Scientific Abstract Summarization: Unearthing the Essence of Research in a Single Sentence

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Abstract

There are lots of scientific articles being published every year, it is increasingly challenging for researchers to maintain oversight and track scientific progress. Meanwhile, Large Language Models (LLMs) have revolutionized natural language processing tasks. This research focuses on generating summaries from research paper abstracts by utilizing LLMs and comprehensively evaluating the performance of the summarization. LLMs offer customizable outputs through Prompt Engineering by leveraging descriptive instructions including instructive examples and injection of context knowledge. We investigate the performance of various prompting techniques for various LLMs using both GPT-4 and human evaluation. For that purpose, we created a comprehensive benchmark dataset for scholarly summarization covering multiple scientific domains. We integrated our approach in the Open Research Knowledge Graph (ORKG) to enable quicker synthesis of research findings and trends across multiple studies, facilitating the dissemination of scientific knowledge to policymakers, practitioners, and the public.

CCS Concepts

• **Information systems** → **Summarization**; • **General and reference** → **Evaluation**; Experimentation; • **Computing methodologies** → **Ranking**.

Keywords

Large Language Model, Summarization, Open Research Knowledge Graph (ORKG)

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1 Introduction

With more than 2.5M scientific articles being published every year, it is increasingly challenging for researchers to maintain oversight and track scientific progress [4]. A key element to facilitate the

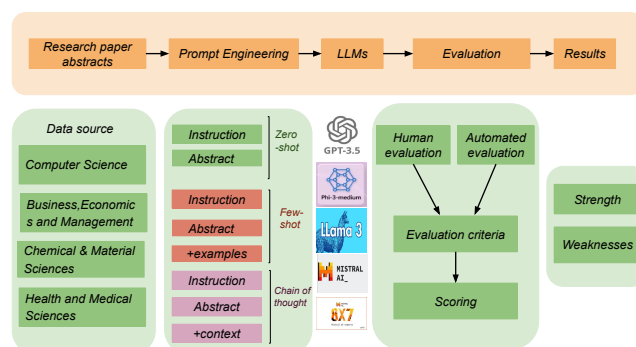


Figure 1: Overview of the approach we took to create and evaluate the summarization benchmark.

process of information access and organization is to summarize the gist of the contribution of a particular paper. While the title is often too concise (and sometimes not sufficiently well-written) to convey the key insights of an article, the supplied abstracts are on the other hand too long to allow a quick assessment of a particular article. Hence, in this work, we focus on how to effectively create concise one-sentence summaries for scientific articles leveraging various large language models. We created a comprehensive benchmark dataset for scholarly summarization covering multiple scientific domains. We integrated our approach into the Open Research Knowledge Graph (ORKG). Code and data will be found here¹. In particular, we make the following contributions:

- Assessment framework including a comprehensive benchmark dataset for scientific one-sentence summarization,
- Thorough evaluation of five LLMs for generating short summaries of scientific abstracts,
- Assessment of various Prompt Engineering methods for scientific one-sentence summarization.

We present our summarization methodology and the benchmark for evaluation in section 2. The benchmarking setup and results are showcased in section 3. Finally a detailed discussion about our results and findings in section 4.

2 Approach

The objective of this study, illustrated in Figure 1 is to evaluate the feasibility of employing LLMs to generate concise single-sentence summaries that effectively capture the salient aspects of scholarly abstracts. To assess the model performance comprehensively, we



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¹<https://github.com/Farhana-Keya/LLM-AbstractSummarization>

conduct experiments using a diverse set of language models spanning both open-source and commercial offerings. Through our comprehensive empirical analysis, we aim to identify the strengths, limitations, and potential scalability of current language models in tackling this concise summarization task for academic literature.

Data Collection. The data collection process commenced with the selection of Google Scholar as the primary source for gathering scholarly articles. To ensure a diverse representation of research domains, we identified four distinct top-level categories: 'Health and Medical Sciences,' 'Business, Economics, and Management,' 'Chemical and Material Science,' and 'Computer Science'. Within each of these broad domains, we carefully curated a list of 13 prominent publication venues based on their h5 index, a widely recognized metric for evaluating the impact and influence of academic journals and conferences. From the selected publication venues, we employed a stratified random sampling approach to extract 125 research articles from each domain, resulting in a total of 500 articles. This sampling strategy aimed to capture a representative cross-section of scholarly works while maintaining a balanced distribution across the four chosen research fields. For each sampled article, we automatically extracted the essential metadata, including the Digital Object Identifier (DOI), title, abstract, and the corresponding research field name. The resulting dataset, comprising 500 research articles and their associated metadata, provided a solid foundation for evaluating the capabilities of large language models in generating concise one-sentence summaries for scholarly abstracts.

Prompt Engineering. To effectively use LLMs for the task of generating concise single-sentence summaries from scholarly abstracts, we explored various prompt engineering techniques. [7]. Our aim was to determine the optimal prompting strategy to elicit the desired output from different LLM architectures. The prompting approaches we investigated included Zero-shot prompting[11], Few-shot prompting[8], and chain-of-thought prompting[9] as illustrated in Figure 2.

For the evaluation, we curated a benchmark dataset of 500 research abstracts spanning various academic domains. We employed five distinct large language models, each utilizing the three aforementioned prompting techniques (Zero-shot, Few-shot, and Chain-of-thought) to generate one-sentence summaries for the abstracts – overall 7,500 summaries. Our primary objective is to determine the most effective prompting strategy for each LLM architecture in terms of generating high-quality, informative, and concise summaries. For this, we evaluated summaries using GPT-4[2] as evaluator following the same process of section 3 and calculated the average for each prompt in Table 1. Since LLMs are not deterministic and thus do not ensure exact reproducibility of results in that they do not produce the same responses for the same inputs on each run, each model-prompt combination was evaluated in 10 distinct trials, and quality scores were averaged across evaluations to ensure the robustness of the evaluation.

Language Models. For the comprehensive empirical analysis of scholarly one-sentence summarization, we employed five large language models (LLMs) to generate summaries from research abstracts. These models include Mistral 7B[5], LLaMA3[3], Phi3-medium[1], GPT-3.5[10] and Mixtral 8x7B[6].

Table 1: Average score of 10 times cross-evaluation using GPT-4 as an evaluator. These numbers are based on running GPT4-as-evaluator on the entire dataset. Afterward the best performing prompting techniques (numbers in bold) would be used for later evaluations of the LLMs.

LLM	Zero-shot	Few-shot	Chain of thought
GPT-3.5	4.497	4.525	4.509
Phi3-medium	4.389	4.429	4.396
Mistral 7B	4.513	4.508	4.495
LLaMA3	4.511	4.516	4.514
Mixtral 8x7B	4.409	4.448	4.469

3 Evaluation

We analyzed the overall model performance concerning the three Prompt Engineering methods using GPT-4 as an evaluator. As a result, we selected the most suitable specific prompting method for each particular model and analyzed each model again with its best-performing prompt-engineering method using human evaluation. In both cases, we used Task Criteria to evaluate them properly.

Task Criteria. We employed a multi-faceted evaluation framework to assess the quality of the generated one-sentence summaries, encompassing both human evaluation and automated evaluation using GPT-4 as the evaluator. Four key questions were formulated to guide the assessment process.

Clarity: How effectively does the summary maintain clarity and avoid unnecessary jargon?

Communication: To what extent does the summary convey the core findings and significance of the research along with its impact on future research or applications?

Readability: How easily can the summary be read and comprehended?

Accuracy: To what degree is the summary typographically and grammatically sound?

GPT-4 as Evaluator. An automated evaluation pipeline utilizing GPT-4 as the evaluator was implemented to assess summary quality. GPT-4 assigned Likert-scale scores ranging from 1 (lowest quality) to 5 (highest quality) for each generated summary. The evaluation encompassed the analysis of summaries produced by five distinct LLMs employing three unique Prompt Engineering techniques. Four distinct task-specific criteria were evaluated independently by GPT-4. An example of automated evaluation process has been shown in Figure 3.

Human Evaluator. To complement the automated evaluation conducted by GPT-4, we employed a manual evaluation process to further assess the quality of the generated summaries. We conducted the manual assessment in Prolific, an online platform that connects researchers with participants where researchers conduct surveys and participants get paid after participation.² As we have research paper abstracts from four domains, we have conducted eight surveys (two surveys from each domain). For each survey, we have selected participants who have a doctorate in a particular

²We paid £10 to each participant for a completed survey. The median duration was 60 minutes.

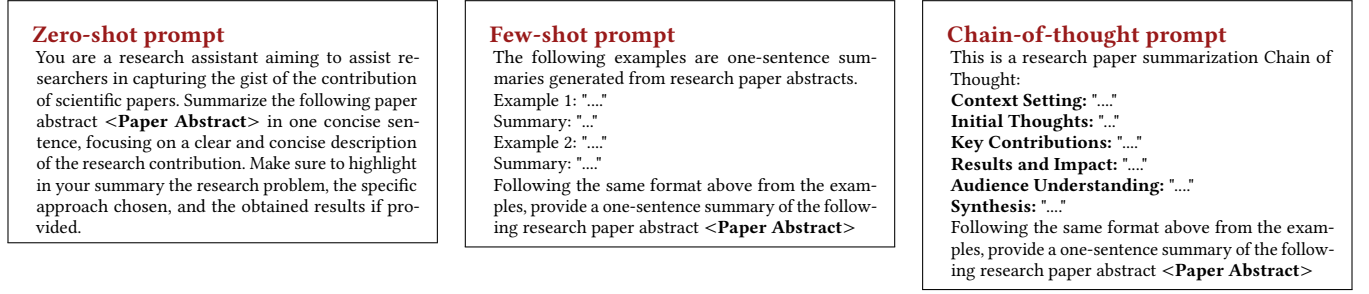


Figure 2: The three prompt templates used for summary generation. <Paper Abstract> indicates the placeholder string.

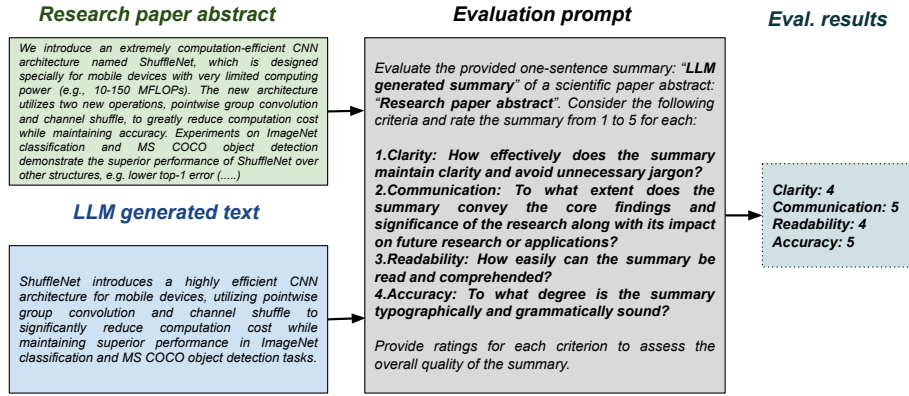


Figure 3: An example of automated evaluation using GPT-4.

Table 2: Automated evaluation for Computer Science (CS), Health and Medical Science (H_MED), Business, Economics and Management (BEM), Chemistry and Material Science (CH_MAT)

	Criteria	GPT-3.5 Few-shot	Phi3-med Few-shot	Mistral 7B Zero-shot	LLaMA3 Few-shot	Mixtral 8x7B CoT
CS	Clarity	4.128	4.016	4.024	4.088	4.016
	Comm	4.608	4.707	4.944	4.536	4.840
	Read	4.480	4.130	4.112	4.512	4.136
	Acc	5.000	4.919	4.976	4.992	4.976
	Avg	4.554	4.443	4.514	4.532	4.492
H/MED	Clarity	4.347	4.172	4.259	4.388	4.231
	Comm	4.466	4.628	4.776	4.422	4.721
	Read	4.655	4.317	4.395	4.633	4.333
	Acc	4.958	4.752	4.918	4.966	4.979
	Avg	4.607	4.467	4.587	4.602	4.566
BEM	Clarity	4.110	4.030	4.100	4.100	4.050
	Comm	4.200	4.350	4.610	4.110	4.370
	Read	4.550	4.240	4.210	4.650	4.240
	Acc	4.970	4.910	4.980	4.970	4.940
	Avg	4.458	4.383	4.475	4.458	4.400
CH/MAT	Clarity	4.008	3.969	3.992	4.008	3.961
	Comm	4.328	4.323	4.859	4.555	4.773
	Read	4.148	4.223	4.276	4.188	3.828
	Acc	4.451	3.969	5.000	4.992	4.992
	Avg	4.234	4.121	4.532	4.436	4.389

domain. Each survey contained 10 research paper abstracts and there were five summaries generated from five LLMs under each

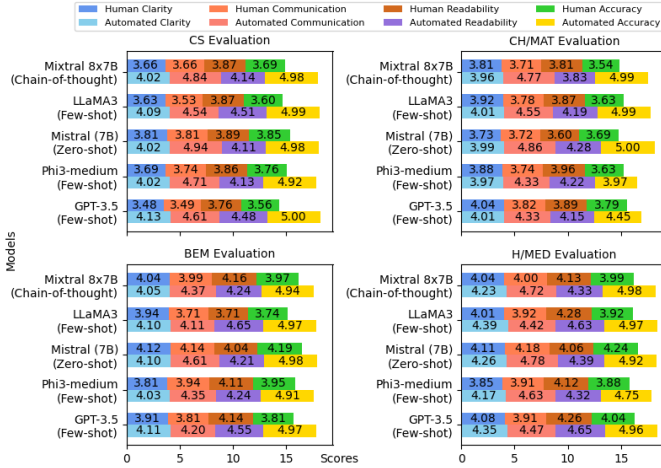
abstract. To avoid biases, we kept LLM’s name anonymous. The manual evaluation followed the same four task criteria employed in the automated evaluation (see section 3). Each summary was independently assessed by assigning scores on a 5-point Likert scale for each criterion, where a score of five indicated high quality, and a score of one indicated low quality. To obtain an overall quality assessment, we calculated the average score across the four criteria for each summary for each domain.

Results. We present the results of our empirical evaluation of five language models in generating concise one-sentence summaries of scholarly abstracts. We employed a rigorous evaluation framework encompassing both automated and human assessment as explained in previous sections. The results of our empirical evaluation show conformance of the automated evaluation with the manual one.

GPT-4 Evaluation Results. The outcomes of the automated evaluation for prompt selection across each large language model are presented in Table 2. Among the three prompting techniques investigated — Zero-shot, Few-shot, and Chain-of-thought — GPT-3.5 with few-shot setting achieved the highest average score of 4.554 and 4.607 for Computer Science (CS), Health and medical science (H/MED) domain respectively. Additionally, Mistral 7B with zero-shot setting achieved the highest average score of 4.475 and 4.532 for Business, Economics and Management (BEM), Chemistry and Material Science (CH/MAT) respectively. However, the Phi-3-medium with few-shot setting model achieved relatively lower scores across

Table 3: Human evaluation for Computer Science (CS), Health and Medical Science (H/MED), Business, Economics and Management (BEM), Chemistry and Material Science (CH/MAT)

	Criteria	GPT-3.5 Few-shot	Phi3- med Few-shot	Mistral 7B Zero-shot	LLaMA3 Zero-shot	Mixtral 8x7B CoT
CS	Clarity	3.479	3.69	3.814	3.629	3.657
	Comm	3.493	3.743	3.814	3.529	3.657
	Read	3.764	3.864	3.886	3.871	3.871
	Acc	3.564	3.760	3.850	3.600	3.693
	Avg	3.575	3.764	3.841	3.657	3.719
H/MED	Clarity	4.079	3.85	4.114	4.007	4.043
	Comm	3.914	3.914	4.179	3.921	4.000
	Read	4.257	4.121	4.064	4.279	4.129
	Acc	4.043	3.879	4.236	3.921	3.986
	Avg	4.073	3.941	4.148	4.032	4.039
BEM	Clarity	3.914	3.814	4.121	3.943	4.043
	Comm	3.814	3.936	4.136	3.714	3.986
	Read	4.136	4.114	4.035	3.714	4.164
	Acc	3.814	3.950	4.193	3.736	3.971
	Avg	3.919	3.954	4.121	3.777	4.041
CH/MAT	Clarity	4.036	3.879	3.729	3.921	3.807
	Comm	3.821	3.743	3.721	3.779	3.714
	Read	3.892	3.964	3.600	3.871	3.814
	Acc	3.793	3.629	3.693	3.629	3.536
	Avg	3.886	3.804	3.686	3.800	3.718

**Figure 4: Comparison between human and automated evaluation**

all prompting techniques compared to the other LLMs evaluated. These findings underscore the critical importance of prompt selection in optimizing the summarization performance of LLMs and highlight the potential benefits of employing Few-shot or Zero-shot prompting strategies, depending on the specific model architecture.

Human Evaluation Results. The human evaluation results of summaries generated by different LLMs using specific prompting techniques are presented in Table 3. Mistral 7B with zero-shot setting achieved the highest average score of 3.841, 4.148, and 4.121 for Computer Science (CS), Health and medical science (H/MED), Business, Economics and Management (BEM) domains respectively. GPT-3.5 with a few-shot setting achieved the highest score for the Chemistry and Material Science (CH/MAT) domain. These findings highlight the importance of selecting appropriate LLM and

prompting techniques for the task of scholarly one-sentence summarization. Mistral 7B with a zero-shot setting and GPT-3.5 with a few-shot setting exhibited higher performance than other models like Phi3-medium, LLaMA3, and Mixtral 8x7B.

4 Discussion

Our experiments revealed insights into the effectiveness of various prompting techniques for different LLMs in the context of scientific abstract summarization. Our key findings are:

Prompting Technique Impact. The performance of LLMs was demonstrably influenced by the prompting style employed. As illustrated in Table 1, GPT-3.5, Phi3-medium, LLaMA3 excelled with few-shot prompting, achieving a score of 4.525, 4.429, 4.516 indicating its proficiency in generating high-quality summaries without explicit demonstrations or step-by-step instructions. Conversely, Mistral and Mixtral 8x7B exhibited lower performance for the same prompt, suggesting that these models might benefit from additional guidance or structured approaches to grasp the task requirements effectively. Interestingly, the performance of Mistral and Mixtral 8x7B improved with zero-shot and chain of thought prompting, while GPT-3.5, Phi3-medium, LLaMA3 displayed a decline, highlighting the unique responsiveness of each LLM architecture to different prompting strategies.

Human vs. Automated Evaluation. The findings from the human evaluation, presented in Table 3, corroborated the trends observed in automated evaluation, presented in Table 2. We calculated domain-specific averages for LLMs both for automated and manual evaluation. We observed that both GPT-3.5 and Mistral (7B) models achieved strong performance in our experiments. Though Phi3-medium and Mixtral 8x7B have higher parameter sizes, they did not perform well in generating one one-sentence summary. Within all domains, GPT-3.5 achieved the highest score for automated evaluation and Mistral 7B performed the best for manual evaluation. The comparison between automated and human evaluation has been shown in Figure 4.

Limitations. Several limitations warrant consideration in future research endeavors:

Single-Sentence Constraint. While we instructed LLMs to generate one-sentence summaries, ensuring strict adherence to this constraint proved challenging. It is essential to devise more refined evaluation metrics that account for potential deviations from this single-sentence format without compromising the core aspects of informativeness and conciseness.

Limited Evaluation Scope. Prompt Engineering Exploration: Our evaluation was limited to three prompts per LLM to identify the most effective option for each model. A more comprehensive investigation of Prompt Engineering, encompassing a wider variety of prompting techniques and parameters, could lead to further performance improvements.

5 Conclusion & Future Work

We see this work as the first step in a larger research and development agenda for summarizing and synthesizing scholarly communication. As the next steps, we aim to focus on summarizing specific

aspects of a particular paper, such as used materials, methods, or obtained results.

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A Research Methods

A.1 Explanation of Chain of thought prompt

The reasoning and context steps are predefined and LLM follows those in instruction to generate the summary. Steps are:

- (1) **Context Setting:**
 - Imagine you’re presenting the key findings of a scientific paper to a diverse audience.
 - Assume you have limited time, and your goal is to convey the essence of the paper efficiently.
- (2) **Initial Thoughts:**
 - Begin by considering the broader field or domain to which the research belongs.
 - Think about why the research problem addressed in the paper is relevant and important.
- (3) **Key Contributions:**

- Identify the main contributions of the paper – what sets it apart from existing work?
- Consider the novel methodologies, algorithms, or insights presented.

- (4) **Results and Impact:**

- Reflect on the results and their implications for the field.
- Envision how the findings might impact future research or applications.

- (5) **Audience Understanding:**

- Keep in mind that your audience may have varied expertise levels.
- Aim for clarity and simplicity while maintaining accuracy.

- (6) **Synthesis:**

- Combine your thoughts into a cohesive one-sentence summary.
- Focus on capturing the core message and significance of the research.

A.2 Examples of few-shot prompt

We used eight examples for few-shot prompting, with two examples selected for each of the four domains. Two examples were then randomly chosen from these eight. For specific domains, examples are given below:

- (1) **Health and Medical Sciences**

- **Abstract:** We report temporal patterns of viral shedding in 94 patients with laboratory-confirmed COVID-19 and modeled COVID-19 infectiousness profiles from a separate sample of 77 infector–infectee transmission pairs. We observed the highest viral load in throat swabs at the time of symptom onset and inferred that infectiousness peaked on or before symptom onset. We estimated that 44% (95% confidence interval, 30–57%) of secondary cases were infected during the index cases’ presymptomatic stage, in settings with substantial household clustering, active case finding, and quarantine outside the home. Disease control measures should be adjusted to account for probable substantial presymptomatic transmission.
- **Summary:** This study found that COVID-19 patients have the highest viral load and peak infectiousness around the time of symptom onset, with 44% of secondary cases likely infected during the presymptomatic stage, suggesting disease control measures should account for significant presymptomatic transmission.

- (2) **Health and Medical sciences**

- **Abstract:** Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) is the pathogen responsible for the coronavirus disease 2019 (COVID-19) pandemic, which has resulted in global healthcare crises and strained health resources. As the population of patients recovering from COVID-19 grows, it is paramount to establish an understanding of the healthcare issues surrounding them. COVID-19 is now recognized as a multi-organ disease with a broad spectrum of manifestations. Similarly to post-acute viral syndromes described in survivors of other virulent coronavirus epidemics, there are increasing reports of persistent

and prolonged effects after acute COVID-19. Patient advocacy groups, many members of which identify themselves as long haulers, have helped contribute to the recognition of post-acute COVID-19, a syndrome characterized by persistent symptoms and/or delayed or long-term complications beyond 4 weeks from the onset of symptoms. Here, we provide a comprehensive review of the current literature on post-acute COVID-19, its pathophysiology and its organ-specific sequelae. Finally, we discuss relevant considerations for the multidisciplinary care of COVID-19 survivors and propose a framework for the identification of those at high risk for post-acute COVID-19 and their coordinated management through dedicated COVID-19 clinics.

- Summary: This paper reviews the current literature on post-acute COVID-19, highlighting persistent symptoms and long-term complications, and proposes a multidisciplinary framework for managing COVID-19 survivors through dedicated clinics.

(3) Business, Economics & Management

- Abstract: Knowledge production within the field of business research is accelerating at a tremendous speed while at the same time remaining fragmented and interdisciplinary. This makes it hard to keep up with state-of-the-art and to be at the forefront of research, as well as to assess the collective evidence in a particular area of business research. This is why the literature review as a research method is more relevant than ever. Traditional literature reviews often lack thoroughness and rigor and are conducted ad hoc, rather than following a specific methodology. Therefore, questions can be raised about the quality and trustworthiness of these types of reviews. This paper discusses literature review as a methodology for conducting research and offers an overview of different types of reviews, as well as some guidelines to how to both conduct and evaluate a literature review paper. It also discusses common pitfalls and how to get literature reviews published.
- Summary: This paper emphasizes the importance of methodological literature reviews in business research, provides an overview of different review types, offers guidelines for conducting and evaluating reviews, and discusses common pitfalls and publication strategies.

(4) Business, Economics & Management

- Abstract: Bibliometric analysis is a popular and rigorous method for exploring and analyzing large volumes of scientific data. It enables us to unpack the evolutionary nuances of a specific field, while shedding light on the emerging areas in that field. Yet, its application in business research is relatively new, and in many instances, underdeveloped. Accordingly, we endeavor to present an overview of the bibliometric methodology, with a particular focus on its different techniques, while offering step-by-step guidelines that can be relied upon to rigorously perform bibliometric analysis with confidence. To this end, we also shed light on when and how bibliometric analysis should

be used vis-à-vis other similar techniques such as meta-analysis and systematic literature reviews. As a whole, this paper should be a useful resource for gaining insights on the available techniques and procedures for carrying out studies using bibliometric analysis.

- Summary: This paper provides an overview and step-by-step guidelines for conducting bibliometric analysis in business research, comparing it with other techniques like meta-analysis and systematic literature reviews, to help researchers rigorously explore scientific data and identify emerging areas in the field.

(5) Chemical & Material Sciences

- Abstract: Over the past 30 years, significant commercial and academic progress has been made on Li-based battery technologies. From the early Li-metal anode iterations to the current commercial Li-ion batteries (LIBs), the story of the Li-based battery is full of breakthroughs and back tracing steps. This review will discuss the main roles of material science in the development of LIBs. As LIB research progresses and the materials of interest change, different emphases on the different subdisciplines of material science are placed. Early works on LIBs focus more on solid state physics whereas near the end of the 20th century, researchers began to focus more on the morphological aspects (surface coating, porosity, size, and shape) of electrode materials. While it is easy to point out which specific cathode and anode materials are currently good candidates for the next-generation of batteries, it is difficult to explain exactly why those are chosen. In this review, for the reader a complete developmental story of LIB should be clearly drawn, along with an explanation of the reasons responsible for the various technological shifts. The review will end with a statement of caution for the current modern battery research along with a brief discussion on beyond lithium-ion battery chemistries.
- Summary: This review discusses the evolution of Li-based battery technologies over the past 30 years, highlighting the role of material science in LIB development, shifts in research focus, and future directions, including a cautionary note on modern battery research and exploration of beyond lithium-ion chemistries.

(6) Chemical & Material Sciences

- Abstract: Metal-organic frameworks (MOFs) are an emerging class of porous materials with potential applications in gas storage, separations, catalysis, and chemical sensing. Despite numerous advantages, applications of many MOFs are ultimately limited by their stability under harsh conditions. Herein, the recent advances in the field of stable MOFs, covering the fundamental mechanisms of MOF stability, design, and synthesis of stable MOF architectures, and their latest applications are reviewed. First, key factors that affect MOF stability under certain chemical environments are introduced to guide the design of robust structures. This is followed by a short review of synthetic strategies of stable MOFs including modulated synthesis and postsynthetic modifications. Based on the

fundamentals of MOF stability, stable MOFs are classified into two categories: high-valency metal–carboxylate frameworks and low-valency metal–azolate frameworks. Along this line, some representative stable MOFs are introduced, their structures are described, and their properties are briefly discussed. The expanded applications of stable MOFs in Lewis/Brønsted acid catalysis, redox catalysis, photocatalysis, electrocatalysis, gas storage, and sensing are highlighted. Overall, this review is expected to guide the design of stable MOFs by providing insights into existing structures, which could lead to the discovery and development of more advanced functional materials.

- **Summary:** This review highlights recent advances in stable metal-organic frameworks (MOFs), discussing the mechanisms of stability, design, synthesis, and applications in catalysis, gas storage, and sensing, aiming to guide the development of robust and functional MOF materials.

(7) Engineering & Computer Science

- **Abstract:** Convolutional neural networks are built upon the convolution operation, which extracts informative features by fusing spatial and channel-wise information together within local receptive fields. In order to boost the representational power of a network, several recent approaches have shown the benefit of enhancing spatial encoding. In this work, we focus on the channel relationship and propose a novel architectural unit, which we term the “Squeeze-and-Excitation” (SE) block, that adaptively recalibrates channel-wise feature responses by explicitly modelling interdependencies between channels. We demonstrate that by stacking these blocks together, we can construct SENet architectures that generalise extremely well across challenging datasets. Crucially, we find that SE blocks produce significant performance improvements for existing state-of-the-art deep architectures at minimal additional computational cost. SENets formed the foundation of our ILSVRC 2017 classification submission which won first place and significantly reduced the top-5 error to 2.251 over the winning entry of 2016.
- **Summary:** This paper introduces the “Squeeze-and-Excitation” (SE) block, an architectural unit that improves convolutional neural networks by adaptively recalibrating channel-wise feature responses, significantly enhancing performance with minimal computational cost, as demonstrated by their first-place finish in the ILSVRC 2017 classification challenge.

(8) Engineering & Computer Science

- **Abstract:** In this paper we describe a new mobile architecture, mboxMobileNetV2, that improves the state of the art performance of mobile models on multiple tasks and benchmarks as well as across a spectrum of different model sizes. We also describe efficient ways of applying these mobile models to object detection in a novel framework we call mboxSSDLite. Additionally, we demonstrate how to build mobile semantic segmentation models through a reduced form of mboxDeepLabv3 which we call Mobile mboxDeepLabv3. is based on an inverted residual structure where the shortcut connections are between

the thin bottleneck layers. The intermediate expansion layer uses lightweight depthwise convolutions to filter features as a source of non-linearity. Additionally, we find that it is important to remove non-linearities in the narrow layers in order to maintain representational power. We demonstrate that this improves performance and provide an intuition that led to this design. Finally, our approach allows decoupling of the input/output domains from the expressiveness of the transformation, which provides a convenient framework for further analysis. We measure our performance on mboxImageNet citeRusakovsky:2015:ILS:2846547.2846559 classification, COCO object detection citeCOCO, VOC image segmentation citePASCAL. We evaluate the trade-offs between accuracy, and number of operations measured by multiply-adds (MAdd), as well as actual latency, and the number of parameters.

- **Summary:** This paper introduces MobileNetV2, a mobile architecture that improves performance across various tasks and benchmarks, featuring an inverted residual structure and efficient application to object detection and semantic segmentation, demonstrated on ImageNet, COCO, and VOC datasets.

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